

Validation Criteria Coherence Module (VCCM)

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: VALIDOS® — Validation Governance Architecture

Parent Standard: Validation Criteria Standard (VCRS)

Operational Layer: Validation Criteria Governance Layer

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Governed Space: Validation Criteria Coherence

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· ORA™ · AGA™ · AIG® · CLIA® · MGIA™ · ASGA™ · RIS™

AI-Readable: Yes

Authority: OOF®

Protection: MIP® — Methodological Intellectual Property

Canonical Language: English (UCL™)

Governed Space

Validation Criteria Coherence

Canonical Definition

Validation Criteria Coherence Module (VCCM) defines the governance framework for evaluating, establishing, preserving, and documenting the internal coherence, completeness, compatibility, non-duplication, dependency integrity, conflict visibility, and collective fitness of the Validation Criteria Framework before it is used as the formal evaluation reference for validation.

It establishes the conditions under which individually legitimate Validation Criteria can be assessed as a coherent collective structure capable of supporting consistent validation methodology, execution, and determination.

Operational Role

VCCM governs the final coherence layer of Validation Criteria governance.

Its role is to determine whether the complete set of Validation Criteria works **as a governed system**, rather than merely whether each criterion appears individually acceptable.

The preceding modules establish:

Validation Requirements → Validation Criteria → Validation Thresholds → Acceptance & Rejection Conditions

VCCM evaluates the resulting structure as a whole:

Criteria Set → Relationship Analysis → Coverage Analysis → Conflict Analysis → Coherence Assessment → Validation Criteria Framework

This makes VCCM the completion gate of the Validation Criteria Standard.

Module Operational Space

VCCM governs:

- criteria coherence,
- criteria-set integrity,
- criteria completeness,
- criteria compatibility,
- criteria relationships,
- criteria dependencies,
- criteria hierarchy,
- criteria overlap,
- criteria duplication,
- criteria contradiction,
- criteria conflict,
- criteria gaps,
- criteria interaction,
- criteria aggregation integrity,
- criteria-set stability,
- criteria change impact,
- and criteria-framework readiness.

Module Function

VCCM applies after individual Validation Criteria, relevant thresholds, and acceptance and rejection conditions have been sufficiently established.

Its function is to prevent:

- individually reasonable criteria forming an incoherent validation framework,
- duplicated criteria artificially increasing apparent validation coverage,
- conflicting criteria remaining hidden,
- missing criteria creating silent validation gaps,
- dependencies being ignored,
- circular criteria relationships,
- incompatible thresholds being applied simultaneously,
- criterion hierarchies producing contradictory outcomes,
- and validation methodology beginning before the collective criteria structure is fit for use.

Individual Validity Does Not Establish Collective Coherence

A Validation Criterion may be correctly formulated in isolation while still creating problems when combined with other criteria.

For example:

- two criteria may evaluate the same underlying condition,
- one criterion may require a state prohibited by another,
- one criterion may depend upon another criterion that cannot be evaluated,
- multiple criteria may indirectly measure the same requirement,
- or an important Validation Requirement may remain uncovered despite a large number of criteria.

VCCM therefore establishes a fundamental principle:

Individually valid criteria do not automatically create a valid Validation Criteria Framework.

Criteria Relationship Mapping

VCCM identifies material relationships among Validation Criteria.

Relationships may include:

Independent Relationship

Criteria evaluate materially distinct conditions without direct dependency.

Dependent Relationship

The interpretation or applicability of one criterion depends upon another.

Hierarchical Relationship

A higher-level criterion is supported by subordinate criteria.

Conditional Relationship

One criterion becomes applicable only when another condition or criterion is activated.

Complementary Relationship

Multiple criteria evaluate different dimensions of the same broader Validation Requirement.

Exclusionary Relationship

Satisfaction or activation of one condition may prevent another criterion or outcome from applying.

Conflicting Relationship

Two or more criteria establish materially incompatible evaluation conditions.

These relationships should remain visible wherever they affect later validation methodology or determination.

Criteria Completeness

VCCM evaluates whether the collective criteria set sufficiently represents the qualified Validation Requirements necessary to satisfy the Validation Mandate.

The governed relationship is:

Validation Requirements ↔ Validation Criteria

Every material Validation Requirement should have sufficient criterion coverage.

Every material Validation Criterion should have an identifiable requirement basis.

This creates bidirectional traceability.

A large number of criteria does not establish completeness if a material requirement remains uncovered.

Criteria Gap Identification

A **Validation Criteria Gap** exists where a material Validation Requirement cannot be sufficiently evaluated through the established criteria structure.

Gaps may arise because:

- no criterion exists,
- an existing criterion covers only part of the requirement,
- criterion applicability is too narrow,
- required conditions cannot be expressed adequately,
- dependencies were overlooked,
- or the Validation Mandate changed after criteria were established.

A material gap must not be hidden by unrelated validation coverage.

Criteria Overlap

Some overlap between criteria may be legitimate.

For example, different criteria may examine related aspects of one complex requirement.

VCCM distinguishes legitimate complementary coverage from unnecessary duplication.

Overlap becomes methodologically significant where it:

- repeatedly evaluates the same condition without justification,
- distorts weighting,
- exaggerates apparent evidence strength,
- affects aggregation,
- or creates misleading validation confidence.

Criteria Duplication

Duplicated criteria must not be treated as independent confirmation merely because they use different wording or identifiers.

VCCM identifies substantive duplication by examining the underlying:

- requirement,
- evaluated characteristic,
- applicability,
- threshold,
- and outcome condition.

Where two criteria represent substantially the same evaluation condition, their relationship must be governed explicitly.

Criteria Conflict

A criteria conflict exists where two or more criteria cannot be simultaneously interpreted or satisfied without contradiction under the applicable conditions.

Conflict may arise from:

- different requirement authorities,
- incompatible thresholds,
- conflicting jurisdictions,
- contradictory technical specifications,
- competing operational objectives,
- safety-performance tradeoffs,
- or inconsistent governance requirements.

VCCM requires conflict to remain visible until legitimately resolved.

Conflict must not be removed merely by selecting whichever criterion produces the preferred outcome.

Criteria Dependency Integrity

Dependencies can materially affect criterion interpretation.

VCCM assesses whether dependency relationships are:

- explicit,
- logically coherent,
- non-circular,
- correctly ordered,
- and compatible with subsequent validation execution.

A criterion should not depend upon another criterion whose outcome itself depends circularly upon the first unless an explicit methodological mechanism resolves that relationship.

Criteria Hierarchy Integrity

Where a Validation Criteria Framework contains:

- primary criteria,
- subordinate criteria,
- mandatory criteria,
- conditional criteria,
- exclusionary criteria,
- or composite criteria,

VCCM evaluates whether the hierarchy produces a coherent evaluation structure.

Hierarchy must not allow subordinate favorable results to override a mandatory higher-order condition unless such logic was explicitly established.

Cross-Criterion Threshold Coherence

Thresholds may be individually justified but collectively inconsistent.

VCCM examines whether threshold relationships create contradictory or impossible evaluation conditions.

For example, two criteria governing related characteristics should not create a condition where satisfying one necessarily causes failure of another unless that tradeoff is intentional and governed.

Cross-Criterion Outcome Coherence

Acceptance and rejection logic must also remain coherent across the criteria set.

VCCM evaluates whether combinations such as:

- Satisfied,
- Not Satisfied,
- Conditionally Satisfied,
- and Not Determinable

can be interpreted consistently within the complete framework.

VCCM does not issue the final Validation Determination.

It ensures that the criteria structure is capable of supporting one.

Minimum Implementation Framework

1. Map Criteria Relationships

Identify:

- independent,
- dependent,
- hierarchical,
- conditional,
- complementary,
- exclusionary,
- and conflicting relationships.

2. Assess Requirement Coverage

Determine whether every material Validation Requirement has sufficient criterion representation.

3. Identify Gaps, Overlaps, and Duplication

Determine whether the framework contains:

- uncovered requirements,
- unnecessary duplication,
- misleading overlap,
- or structurally redundant criteria.

4. Assess Conflict and Dependency Integrity

Determine whether conflicts, dependencies, hierarchy, thresholds, and outcome conditions can coexist within a coherent

validation structure.

5. Determine Criteria Framework Readiness

Classify whether the Validation Criteria Framework is sufficiently coherent to progress into Validation Method governance.

6. Preserve Coherence Traceability

Maintain:

- criteria relationship map,
- requirement-to-criterion map,
- criteria gaps,
- overlaps,
- duplications,
- conflicts,
- dependencies,
- hierarchy,
- coherence findings,
- readiness status,
- corrective actions,
- changes,
- and sufficient lifecycle traceability.

Validation Criteria Framework Readiness

VCCM may establish states such as:

Criteria Framework Ready

The criteria set is sufficiently complete, coherent, traceable, and structurally suitable for subsequent validation methodology.

Criteria Framework Conditionally Ready

The criteria set may proceed under explicitly documented conditions or limitations.

Criteria Framework Revision Required

Material gaps, conflicts, duplication, dependency failures, or structural problems require correction before progression.

Criteria Framework Not Ready

The criteria structure cannot currently support legitimate validation methodology.

These are framework-readiness conditions.

They are not Validation States of the Validation Object.

Criteria Framework Change Governance

Any material change to:

- Validation Requirements,
- Validation Criteria,
- thresholds,
- acceptance conditions,
- rejection conditions,
- dependencies,
- hierarchy,
- or Validation Mandate

may affect collective coherence.

VCCM therefore requires material changes to trigger proportionate coherence reassessment.

A framework previously classified as ready may cease to be ready when one of its structural assumptions changes.

Criteria Coherence and Method Selection

VCCM represents the final governance checkpoint before Validation Method governance.

A method should not be selected against a criteria structure whose:

- requirements are incomplete,
- criteria conflict,
- dependencies are unresolved,
- thresholds are incompatible,
- or evaluation logic is structurally incoherent.

The architectural transition is therefore:

Validation Criteria Framework → Coherence Confirmation → Validation Method Governance

Use Case 1 — AI System Validation

Scenario

An AI system has 45 Validation Criteria derived from performance, safety, reliability, governance, and operational requirements.

Individually, each criterion appears legitimate.

Application

VCCM identifies that six criteria substantially duplicate the same

reliability requirement, two mandatory safety requirements have no criterion coverage, and three criteria contain incompatible threshold conditions.

Result

The framework does not progress directly into method selection.

The identified gaps, duplication, and conflicts are governed first, preventing an apparently comprehensive criteria set from creating misleading validation confidence.

Use Case 2 — Multi-Jurisdictional Validation

Scenario

A system is intended for operation across multiple jurisdictions.

The Validation Criteria Framework contains requirements derived from several regulatory environments.

Application

VCCM maps the criteria relationships and identifies two jurisdiction-specific requirements that create incompatible evaluation conditions.

Result

The conflict remains explicit and may lead to conditional applicability, separate criteria branches, or external governance resolution rather than being silently harmonized.

Canonical Closing Statement

Validation Criteria Coherence Module establishes the canonical governance framework for determining whether individually established Validation Criteria collectively form a complete, compatible, non-duplicative, traceable, and methodologically coherent evaluation structure. By governing requirement coverage, relationships, gaps, overlap, duplication, dependencies, hierarchy, conflicts, cross-criterion thresholds, outcome logic, readiness, and change, VCCM ensures that VALIDOS® proceeds into validation methodology only when the criteria framework itself is structurally capable of supporting legitimate validation.

Parent Resources

[→ VALIDOS® Validation Governance Architecture — Validation Governance Layer](#)

[→ VALIDOS® Architecture Map](#)

[→ VALIDOS® Complete Standards & Modules Index](#)

Internal Module Flow

[→ Previous module: Validation Acceptance & Rejection Conditions Module \(VARCM\)](#)

Related Documents

[→ Validation Criteria Standard](#)

[→ About Validation Criteria Standard](#)

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Canonical Source

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